**ANGLO-CHINESE JUNIOR COLLEGE**

**JC2 Preliminary Examination**

**Higher 2**

# H2 GEOGRAPHY

**9173/02**

Paper 2

**28 August 2024**

Insert

**3 hours**

**READ THESE INSTRUCTIONS FIRST**

This Insert contains all the Resources referred to in the questions.

This document consists of **9** printed pages and **1** blank page.

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## Resource 1 for Question 1

## Land use map of the area adjacent to the Punggol Waterway




**Scale:** 200 m

Map Legend

| | | |
|---|---|---|
| Open Space | E Educational Institution | Commercial & Residential |
| Park | W Place of Worship | Commercial |
| Beach Area | Civic & Community Institution | Business Park |
| Sports & Recreation | Residential | Transport Facilities |
| Waterbody | Reserve Site | |

## Resource 2 for Question 1

## Photographs of Sites X and Y

The placement of the infiltrometer at each site is marked by a white cylinder in the photographs.

**Site X**

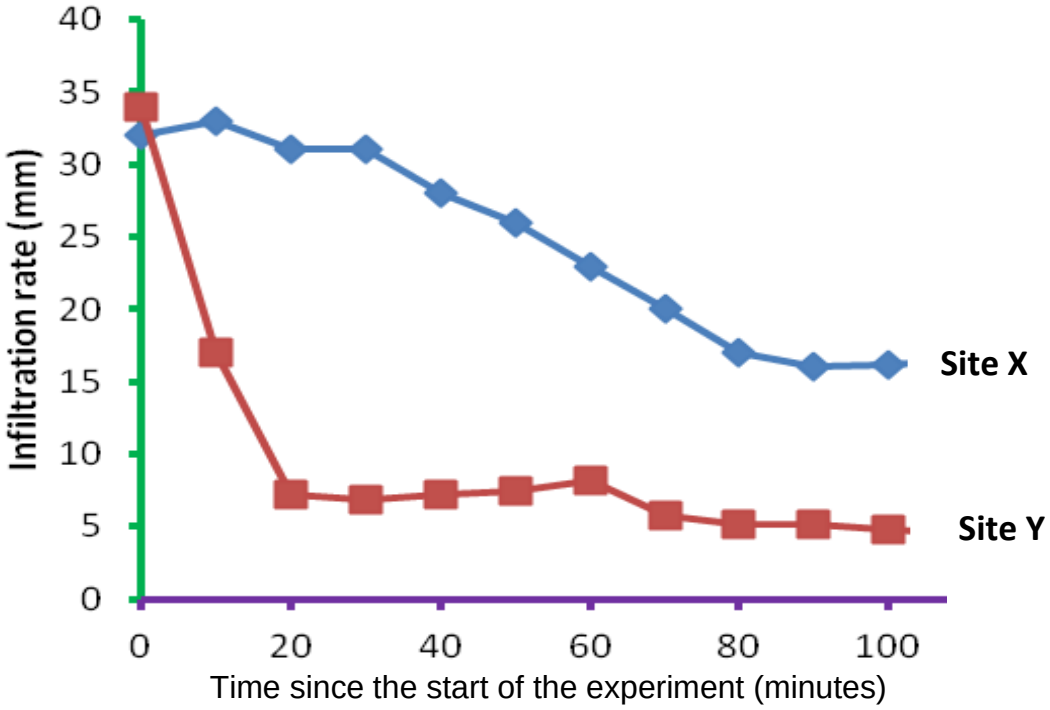


**Site Y**



**Resource 3 for Question 1**

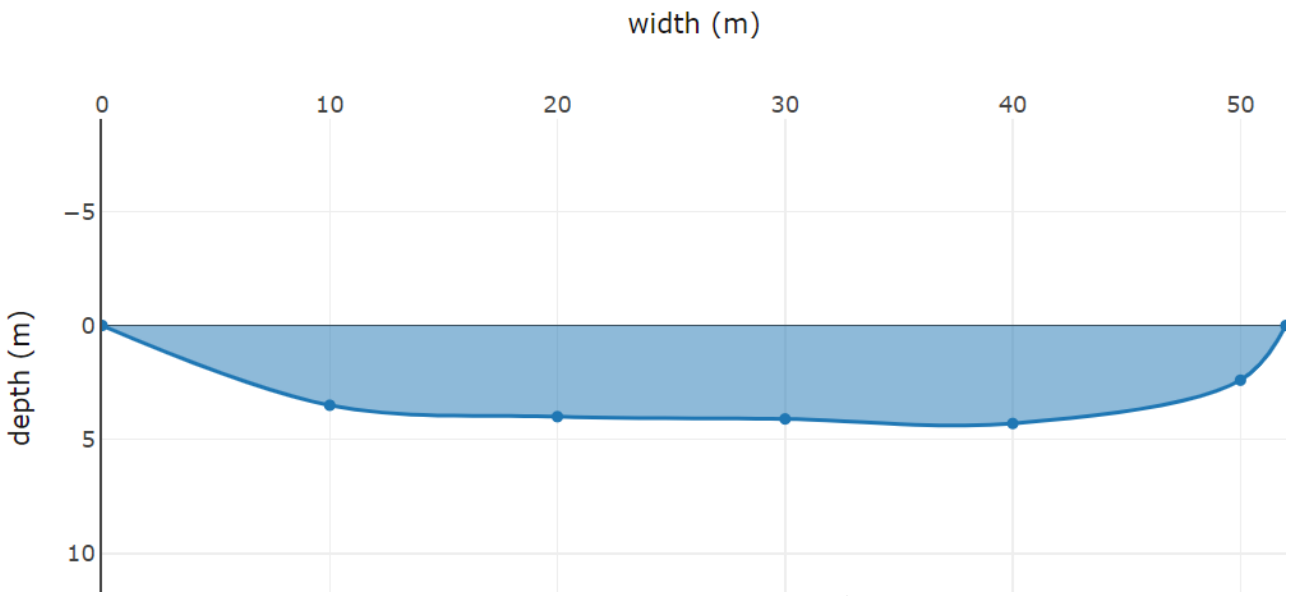
**Infiltration rates at Site X and Site Y**



## Resource 4 for Question 1

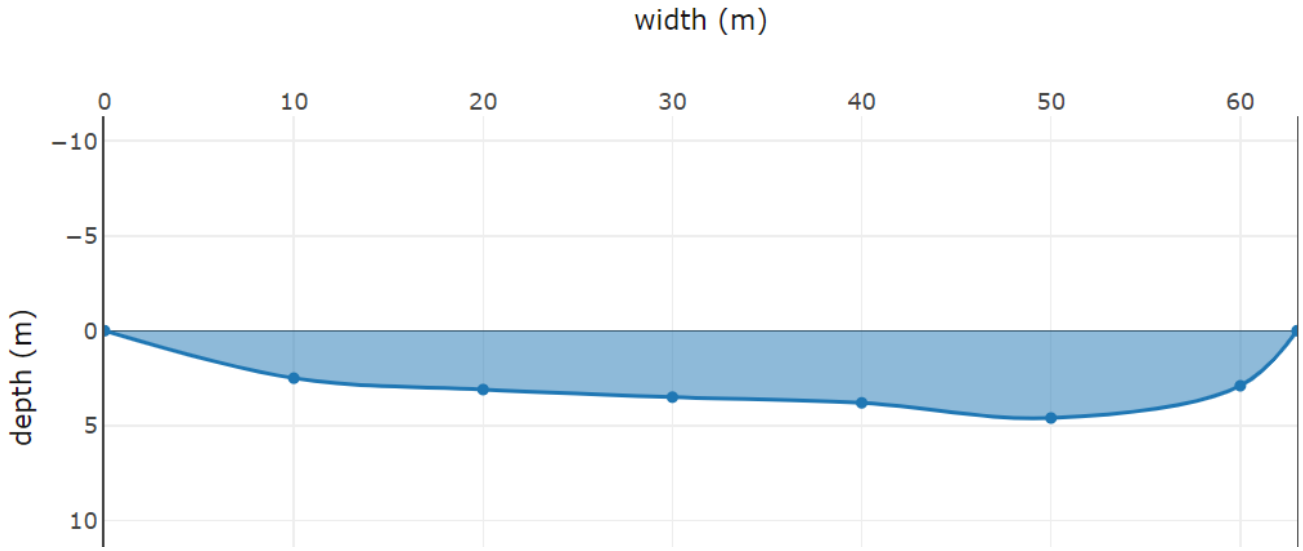
## Results of the river investigation

### River cross section at Site X




Cross sectional area: 173.4 $m^2$
Wetted perimeter: 53.9 m
Hydraulic radius: 3.22
River velocity: 0.082 metres/sec

### River variables at Site Y



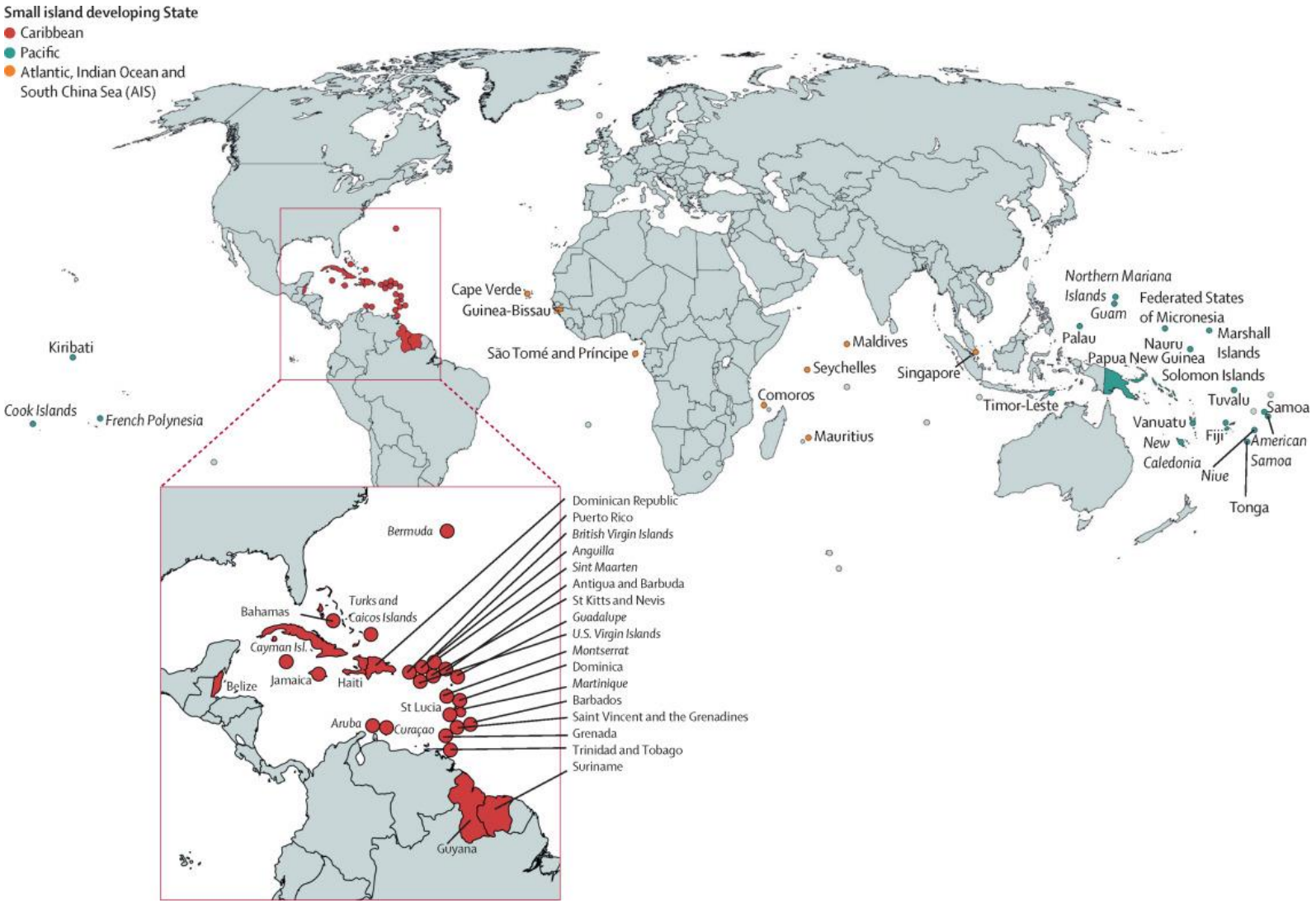

Cross sectional area: 193.8 $m^2$
Wetted perimeter: 67.6 m
Hydraulic radius: 2.9
River velocity: 0.071 metres/sec

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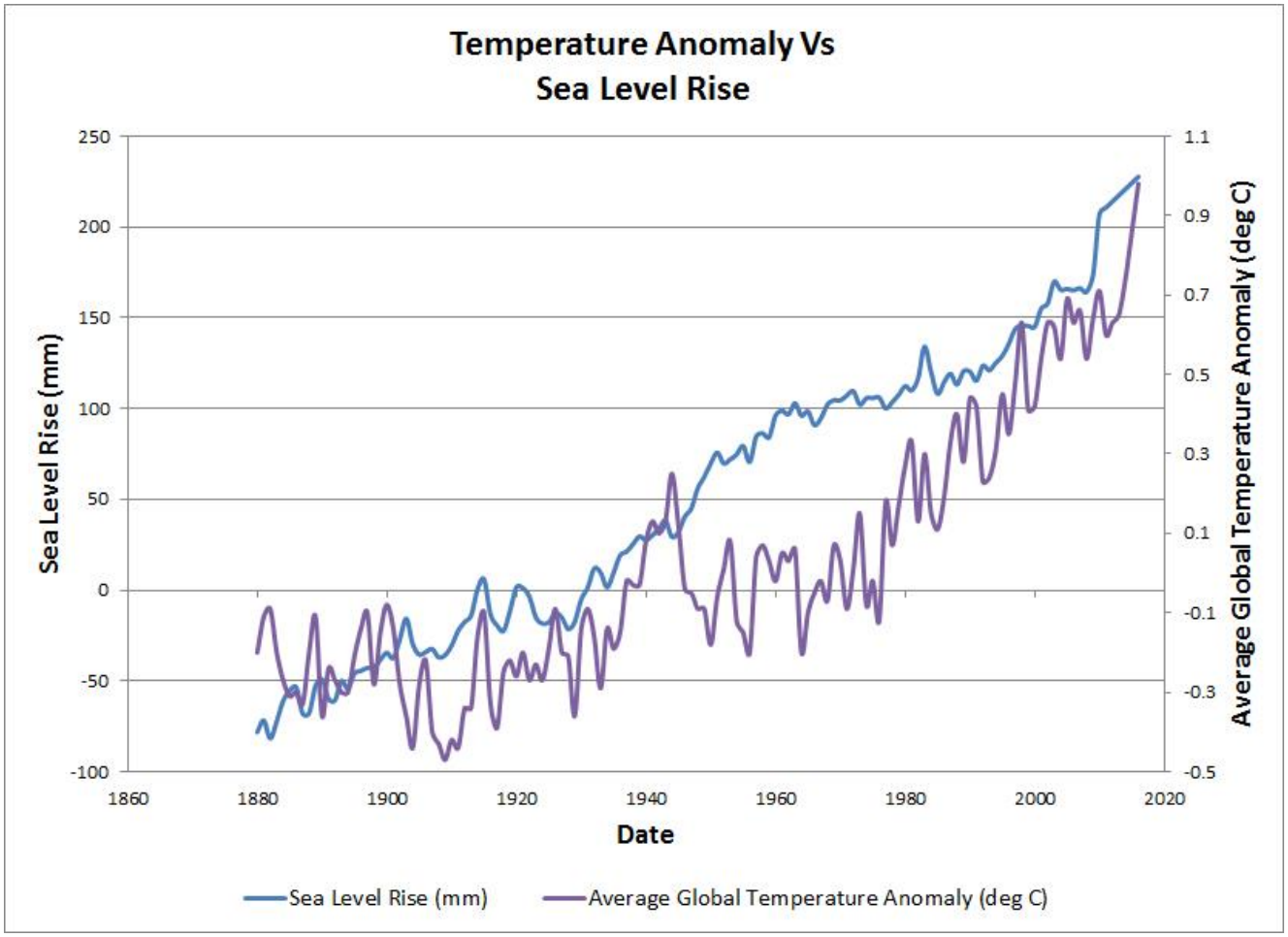
## Resource 5 for Question 2

## Distribution of the Small Island Developing States (SIDS)



## Resource 6 for Question 2

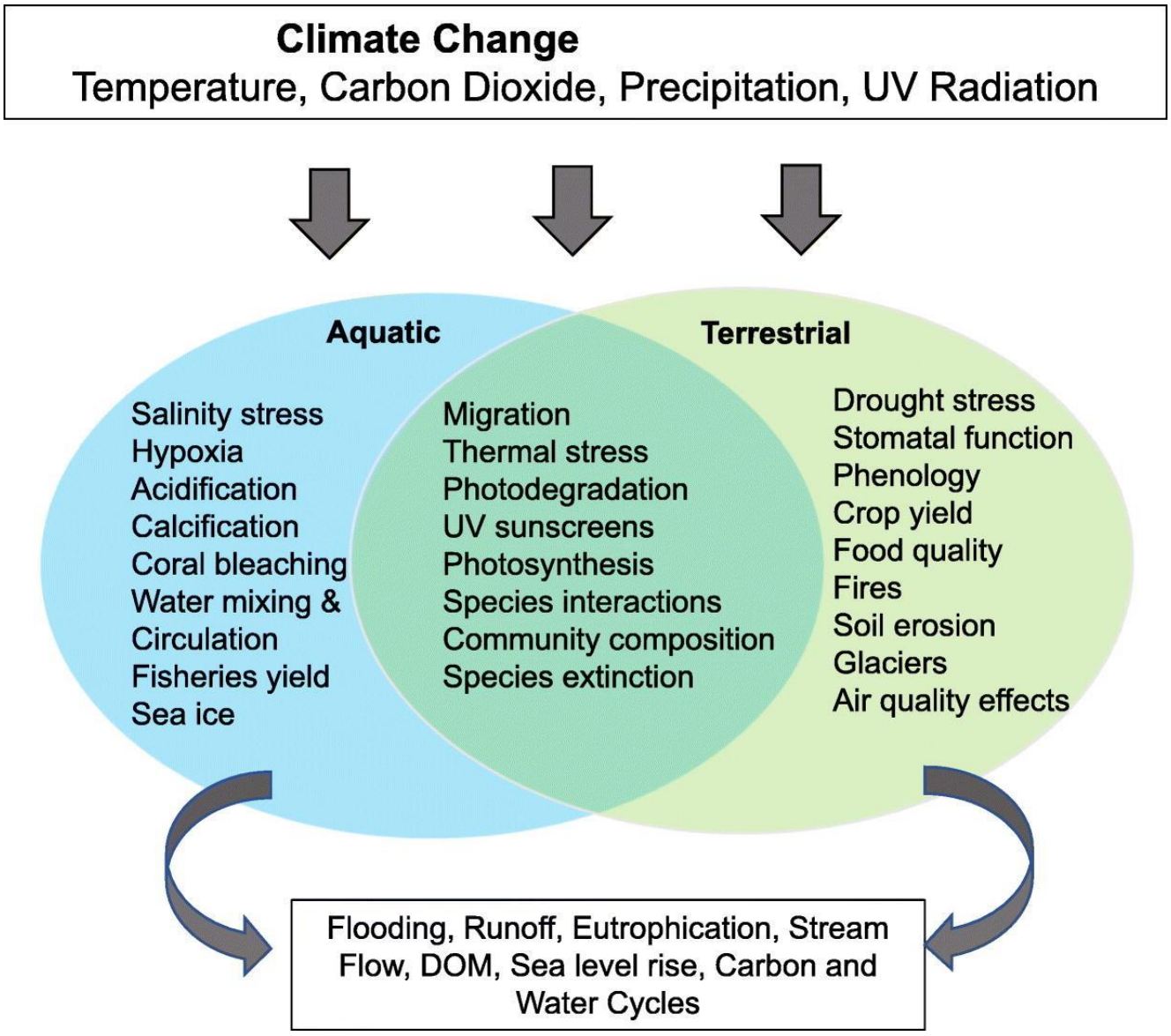
### Temperature anomaly* vs sea level rise




***Temperature anomaly** means a departure from a reference value or long-term average. A positive anomaly indicates that the observed temperature was warmer than the reference value, while a negative anomaly indicates that the observed temperature was cooler than the reference value.

**Resource 7 for Question 2**

**The impacts of climate change on aquatic and terrestrial ecosystems**



**Resource 8 for Question 2**

**Infographic about the impacts of climate change on SIDS**

# Impact of climate change on Small Island Developing States

## Over 3 Billion

people live in areas that are highly susceptible to climate risks, according to the IPCC. This includes most small island developing states.



**Small Island Developing States (SIDS) are vulnerable to the impacts of climate change due to specific geographical conditions.**

**For example:**

- **Small landmasses** (ex: Mauritius, Barbados)
- **Low-lying coastal areas** (ex: Maldives, Tuvalu)
- **Remote locations** (ex: Fiji, Seychelles)
- **Fragile ecosystems** (ex: Palau, the Marshall Islands)
- **Dependence on natural resources** (ex: Dominica, Solomon Islands)
- **Pronounced vulnerability to natural disasters** (Saint Lucia, Antigua and Barbuda)

**Such countries are at high risk of experiencing severe impacts of climate change, including:**

**Sea Level Rise, Ocean acidification and extreme weather events such as:**

- tropical cyclones
- hurricanes
- droughts
- coastal erosion

**Assessing the climate crisis:**

1. 

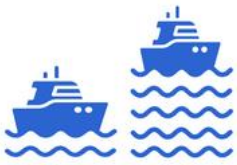
*6* out of the *10* nations at highest risk to climate change and natural disasters worldwide are in *Asia and the Pacific*, 3 of which are tiny Pacific island states.

2. 

In *Mauritius*, without intervention, beach degradation leading to a decline in tourism could cause an estimated annual *loss of over US$100 million* by 2060.

3. 

As islands are threatened to sink soon, countries like *Kiribati* are purchasing land in other countries to *relocate their endangered population*.

4. 

Almost *30%* of the people in SIDS live in areas that are *5 metres or less above sea level* making them extremally vulnerable to climate change.

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